

Toward Aligning Human and Robot Actions via Multi-Modal Demonstration Learning

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Introduction

Motivation:

- 2D Human demos (videos / images) & 3D robots perception (RGB-D/point clouds) create a spatial mismatch.
- Task-specific LfD methods limit scalability and generalization.
- Lack of clear alignment between human intent and robot actions hinders effective imitation.

Contributions:

- **An unified cross-modal framework** aligning 2D human actions with 3D voxelized RGB-D robot inputs.
- **A dual-branch model** for sequential human intent (ResNet18+LSTM) and robot action (Perceiver Transformer).
- Achieves over 71% accuracy on RH20T (8 class tasks) for both intent and action prediction.

Problem Formulation

Objective:

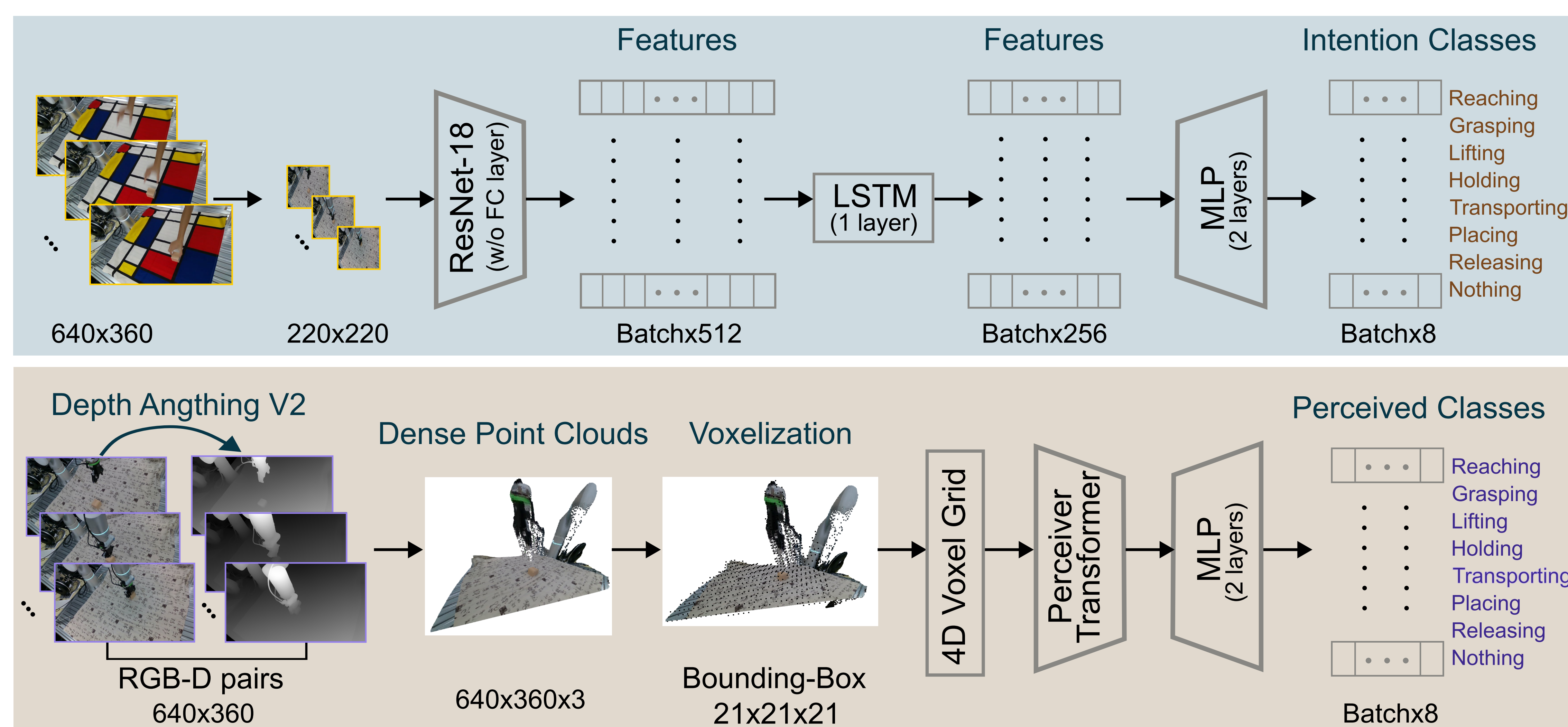
- Introduce a generalized framework to align human-robot actions across visual and geometric modalities.
- Quantify semantic correspondence by defining an alignment score :

$$S(H, R) = \frac{1}{T} \sum_{t=1}^T \delta(i_t, j_t) P(i_t|H) P(j_t|R)$$

- Maximizing this alignment score with respect to model parameters θ :

$$\theta^* = \arg \max_{\theta} S(H, R; \theta).$$

Methodology



Arxiv



Website



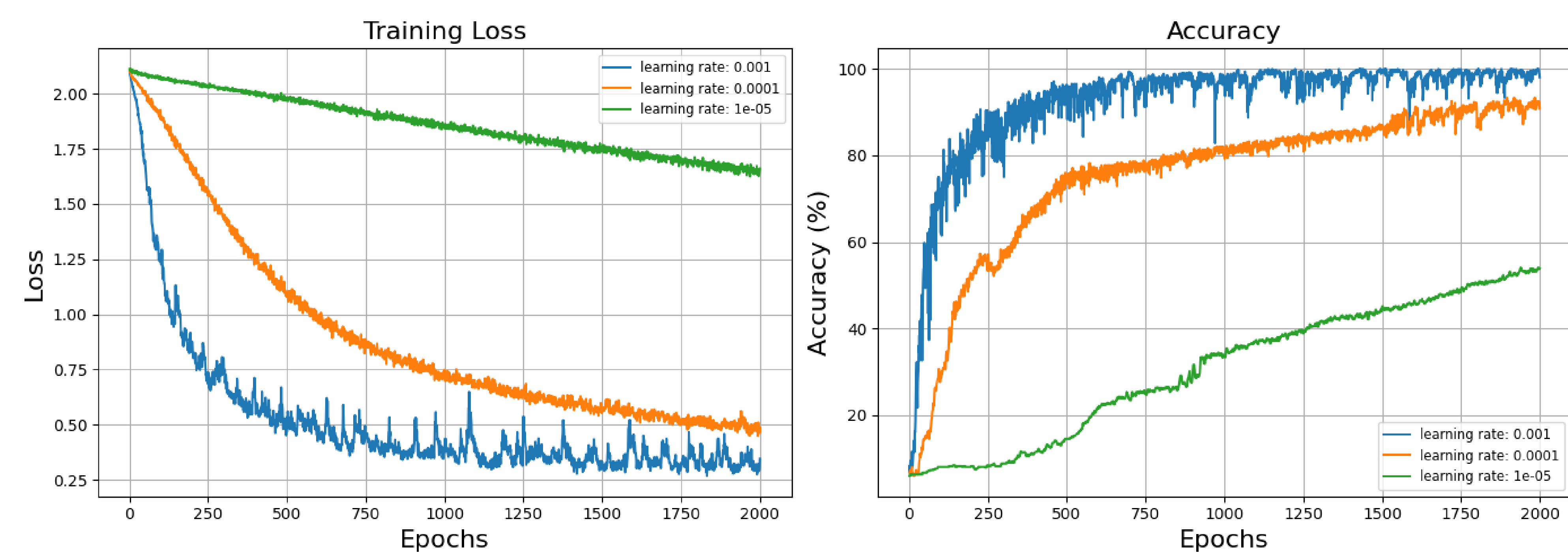
Github

Experimental Results

Trained independently on RH20T (“PICK” task: 5 users, 10 scenes, 8 classes) using weighted cross-entropy.

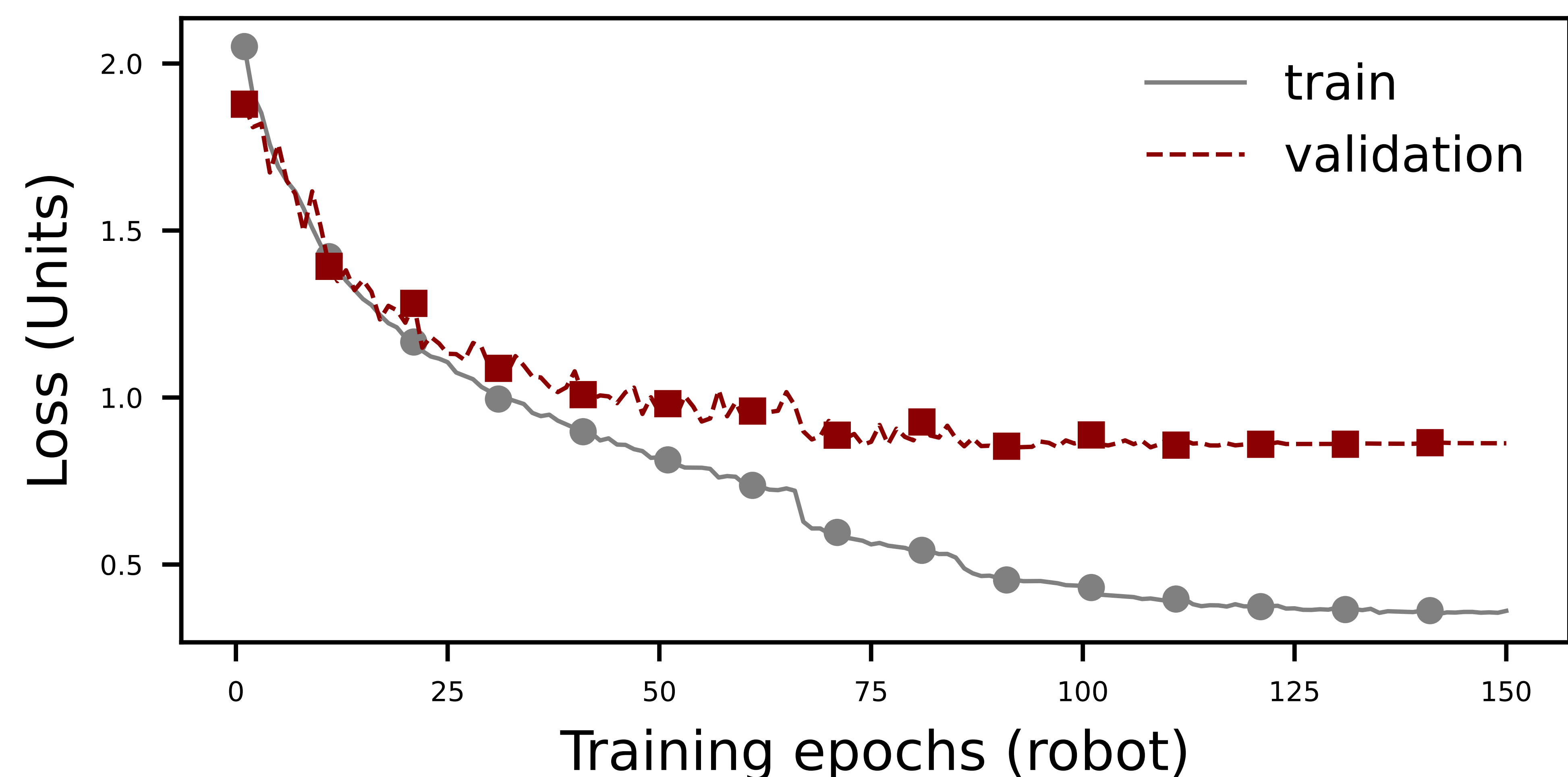
A. Human Branch:

- Validation: **71.7%**



B. Robot Branch:

- Perceiver: **76.0%**
- Prediction: **83.0%**



C. Comparison:

- **Higher confidence:** nothing, holding
- **Lower confidence:** grasping, placing

		Human								Robot							
True label	reaching	24	3	0	0	0	0	0	10	497	50	37	17	41	13	5	74
	grasping	7	3	3	0	0	0	0	0	15	67	5	0	1	2	0	2
	lifting	2	12	4	4	1	0	0	1	11	12	34	2	4	2	3	0
	holding	0	2	5	10	0	0	0	0	8	0	7	110	22	3	3	14
	transporting	0	0	0	0	5	2	0	1	26	3	5	18	142	9	6	11
	placing	0	0	0	0	0	5	1	1	5	0	1	1	4	52	5	2
	releasing	0	0	0	0	0	0	1	2	1	0	0	2	1	6	5	3
	nothing	27	0	0	0	0	0	0	97	76	5	4	17	42	17	11	543
		reaching	grasping	lifting	holding	transporting	placing	releasing	nothing	reaching	grasping	lifting	holding	transporting	placing	releasing	nothing
		Predicted label								Predicted label							